

SYSTEMS ADMINISTRATION USING CISCO® (315)

NATIONAL – 2019

TOTAL POINTS _____ *(500 points)*

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
- 3. Electronic devices will be monitored according to ACT standards.**

No more than sixty (60) minutes testing time

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Identify the letter of the choice that best completes the statement or answers the question.

1. Which command allows you to verify the encapsulation type (CISCO or IETF) for a Frame Relay link?
 - A. show frame-relay lmi
 - B. show frame-relay map
 - C. show frame-relay pvc
 - D. show interfaces serial
2. Which item represents the standard IP ACL?
 - A. access-list 110 permit ip any
 - B. access-list 50 deny 192.168.1.1 0.0.0.255
 - C. access list 101 deny tcp any host 192.168.1.1
 - D. access-list 2500 deny tcp any host 192.168.1.1 eq 22
3. Which protocol is an open standard protocol framework that is commonly used in VPNs, to provide secure end-to-end communications?
 - A. RSA
 - B. L2TP
 - C. IPsec
 - D. PPTP
4. Refer to the exhibit. Statements A, B, C, and D of ACL 10 have been entered in the shown order and applied to interface E0 inbound, to prevent all hosts (except those whose addresses are the first and last IP of subnet 172.21.1.128/28) from accessing the network. But as is, the ACL does *not* restrict anyone from the network. How can the ACL be re-arranged so that the system works as intended?

ACL 10
Statements are written in this order:

- A. permit any
- B. deny 172.21.1.128 0.0.0.15
- C. permit 172.21.1.129 0.0.0.0
- D. permit 172.21.1.142 0.0.0.0

- A. ACDB
- B. BADC
- C. DBAC
- D. CDBA

5. Refer to the exhibit. Why has this switch *not* been elected the root bridge for VLAN1?

```
Switch# show spanning-tree vlan 1
VLAN0001
  Spanning tree enabled protocol rstp
    Root ID    Priority    20481
               Address     0008.217a.5800
               Cost        38
               Port        1 (FastEthernet0/1)
               Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec

    Bridge ID   Priority    32769 (priority 32768 sys-id-ext 1)
               Address     0008.205e.6600
               Hello Time   2 sec   Max Age 20 sec   Forward Delay 15 sec
               Aging Time   300

Interface      Role Sts Cost      Prio.Mbr Type
-----
Fa0/1          Root FWD 19        128.1   P2p
Fa0/4          Desg FWD 38        128.1   P2p
Fa0/11         Altn BLK 57        128.1   P2p
Fa0/13         Desg FWD 38        128.1   P2p
```

- A. It has more than one interface that is connected to the root network segment
 - B. It is running RSTP while the elected root bridge is running 802.1d spanning tree
 - C. It has a higher MAC address than the elected root bridge
 - D. It has a higher bridge ID than the elected root bridge
6. What command sequence will configure a router to run OSPF and add network 10.1.1.0/24 to area 0?
- A. router ospf area 0
network 10.1.1.0 255.255.255.0 area 0
 - B. router ospf
network 10.1.1.0 0.0.0.255
 - C. router ospf 1
network 10.1.1.0 0.0.0.255 area 0
 - D. router ospf area 0
network 10.1.1.0 0.0.0.255 area 0
7. What is the default maximum number of equal-cost paths that can be placed into the routing table of a Cisco OSPF router?
- A. 2
 - B. 8
 - C. 16
 - D. unlimited

8. Which statement about access lists that are applied to an interface is true?
- A. You can place as many access lists as you want on any interface.
 - B. You can apply only one access list on any interface
 - C. You can configure one access list, per direction, per Layer 3 protocol.
 - D. You can apply multiple access lists with the same protocol or in different directions
9. What is one benefit of PVST+?
- A. PVST+ supports Layer 3 load balancing without loops
 - B. PVST+ reduces the CPU cycles for all the switches in the network
 - C. PVST+ allows the root switch location to be optimized per VLAN
 - D. PVST+ automatically selects the root bridge location, to provide optimized bandwidth usage.
10. Refer to the exhibit. Which statement is true?

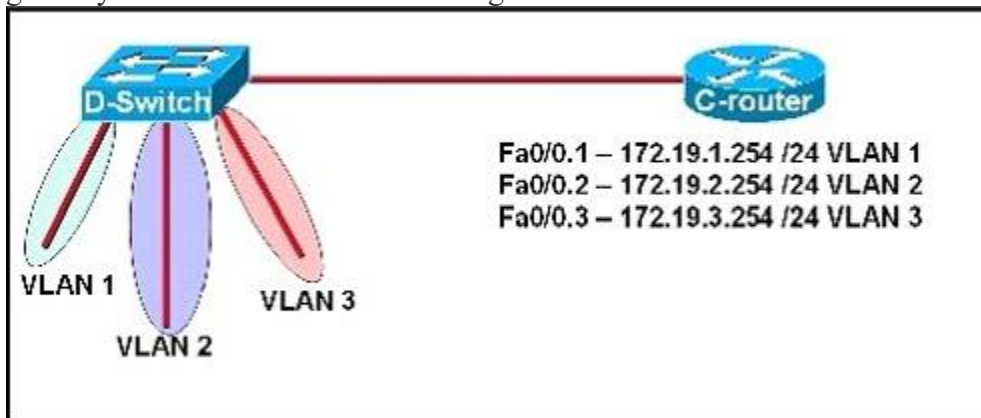
```
SwitchA# show spanning-tree vlan 20

VLAN0020
  Spanning tree enabled protocol rstp
    Root ID    Priority    24596
              Address     0017.596d.2a00
              Cost        38
              Port        11 (FastEthernet0/11)
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
    Bridge ID  Priority    28692 (priority 28672 sys-id-ext 20)
              Address     0017.596d.1580
              Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
              Aging Time  300

Interface      Role Sts Cost      Prio.Nbr Type
-----
Fa0/11         Root FWD 19       128.11  P2p
Fa0/12         Altn BLK 19       128.12  P2p
```

- A. The Fa0/11 role confirms that SwitchA is the root bridge for VLAN 20.
- B. VLAN 20 is running the Per VLAN Spanning Tree Protocol
- C. The MAC address of the root bridge is 0017.596d.1580.
- D. SwitchA is not the root bridge, because not all of the interface roles are designated.

11. Refer to the exhibit. C-router is to be used as a “router-on-a-stick” to route between the VLANs. All the interfaces have been properly configured and IP routing is operational. The hosts in the VLANs have been configured with the appropriate default gateway. What is true about this configuration?



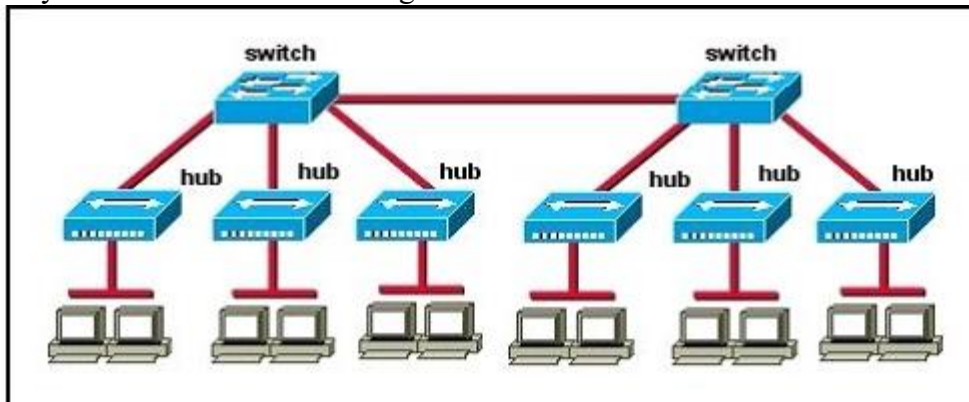
- A. These commands need to be added to the configuration:
C-router(config)#router eigrp 123
C-router(config-router)#network 172.19.0.0
 - B. These commands need to be added to the configuration:
C-router(config)#router ospf 1
C-router(config-router)#network 172.19.0.0 0.0.3.255 area 0
 - C. These commands need to be added to the configuration:
C-router(config)#router rip
C-router(config-router)#network 172.19.0.0
 - D. No further routing configuration is required
12. Which type of device can be replaced by the use of subinterfaces for VLAN routing?
- A. Layer 2 bridge
 - B. Layer 2 switch
 - C. Layer 3 switch
 - D. router
13. Which type of EIGRP route entry describes a feasible successor?
- A. a backup route, stored in the routing table
 - B. a primary route, stored in the routing table
 - C. a backup route, stored in the topology table
 - D. a primary route, stored in the topology table
14. When you are troubleshooting an ACL issue on a router, which command would you use to verify which interfaces are affected by the ACL?
- A. show ip access-lists
 - B. show access-lists
 - C. show interface
 - D. show ip interface

15. Which statement describes the process ID that is used to run OSPF on a router?
- A. It is globally significant and is used to represent the AS number
 - B. It is locally significant and is used to identify an instance of the OSPF database
 - C. It is globally significant and is used to identify OSPF stub areas.
 - D. It is locally significant and must be the same throughout an area

16. Refer to the exhibit. What is the effect of the given configuration?

```
CiscoSwitch-MDF-1#configure terminal
CiscoSwitch-MDF-1#interface VLAN 1
CiscoSwitch-MDF-1(config-if)#ip address 192.168.2.2 255.255.255.0
CiscoSwitch-MDF-1(config-if)#end
```

- A. It configures an inactive switch virtual interface.
 - B. It configures an active management interface.
 - C. It configures the native VLAN.
 - D. It configures the default VLAN.
17. What occurs on a Frame Relay network when the CIR is exceeded?
- A. All TCP traffic is marked discard eligible.
 - B. All UDP traffic is marked discard eligible and a BECN is sent.
 - C. All TCP traffic is marked discard eligible and a BECN is sent.
 - D. All traffic exceeding the CIR is marked discard eligible.
18. Refer to the exhibit. How many broadcast domains are shown in the graphic assuming only the default VLAN is configured on the switches?



- A. one
- B. two
- C. six
- D. twelve

19. A router has two Fast Ethernet interfaces and needs to connect to four VLANs in the local network. How can you accomplish this task, using the fewest physical interfaces and without decreasing network performance?
- A. Use a hub to connect the four VLANs with a Fast Ethernet interface on the router.
 - B. Add a second router to handle the VLAN traffic.
 - C. Add two more Fast Ethernet interfaces.
 - D. Implement a router-on-a-stick configuration.
20. What is the default administrative distance of OSPF?
- A. 90
 - B. 100
 - C. 110
 - D. 120
21. Which benefit is provided by creating VLANs?
- A. dedicated bandwidth
 - B. added security
 - C. allows switches to route traffic between subinterfaces
 - D. contains collisions
22. Which spanning-tree protocol rides on top of another spanning-tree protocol?
- A. MSTP
 - B. RSTP
 - C. PVST+
 - D. Mono Spanning Tree
23. A network administrator is trying to add a new router into an established OSPF network. The networks attached to the new router do *not* appear in the routing tables of the other OSPF routers. Given the information in the partial configuration show below, what configuration error is causing this problem?
- ```
Router(config)#router ospf 1
Router(config-router)#network 10.0.0.0 255.0.0.0 area 0
```
- A. The process id is configured improperly.
  - B. The OSPF area is configured improperly.
  - C. The network wildcard mask is configured improperly.
  - D. The AS is configured improperly.

24. What additional command is required to properly configure the router to add network 192.168.16.0/24 to OSPF area 0?

Router(config)#router ospf 1

- A. Router(config-router)#network 10.0.0.0 255.0.0.0 area 0
- B. Router(config-router)#network 192.168.16.0 0.0.0.255 area 0
- C. Router(config-router)#network 192.168.16.0 0.0.0.255 area 0
- D. Router(config-router)#network 192.168.16.0 255.255.255.0 area 0

25. Which statement describes VRRP object tracking?

- A. It monitors traffic flow and link utilization.
- B. It ensures the best VRRP router is the virtual router master for the group.
- C. It causes traffic to dynamically move to higher bandwidth links.
- D. It thwarts man-in-the-middle attacks.

26. Which port state is introduced by Rapid-PVST?

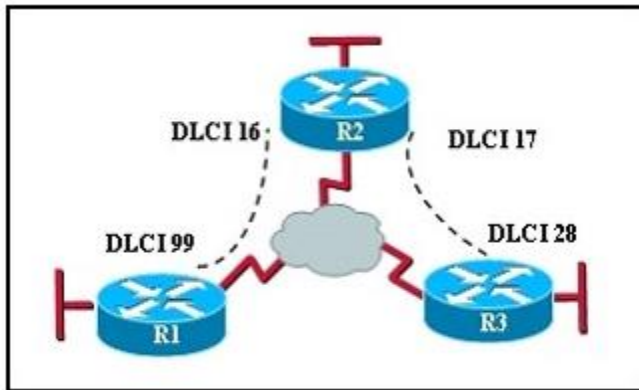
- A. learning
- B. listening
- C. discarding
- D. forwarding

27. Refer to the exhibit. What is the meaning of the term dynamic as displayed in the output of the show frame-relay map command shown?

```
R1# show frame-relay map
Serial0/0 (up): ip 172.16.3.1 dlci 100 (0x64, 0x1840), dynamic
 broadcast, status defined, active
```

- A. The mapping between DLCI 100 and the end station IP address 172.16.3.1 was learned through Inverse ARP.
- B. The Serial0/0 interface is passing traffic.
- C. The DLCI 100 was dynamically allocated by the router.
- D. The DLCI 100 will be dynamically changed as required to adapt to changes in the Frame Relay cloud.

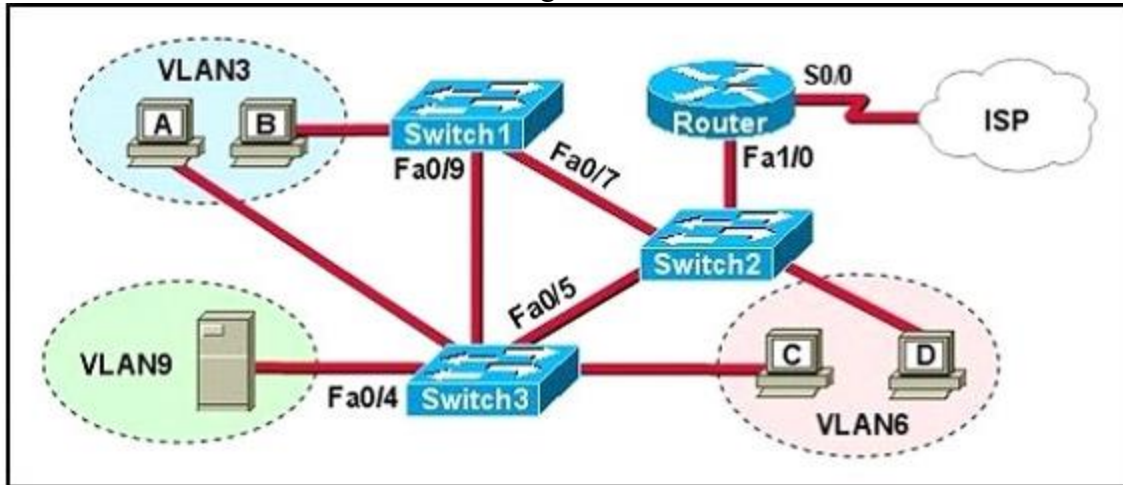
28. Refer to the exhibit. In the Frame Relay network, which IP addresses would be assigned to the interfaces with point-to-point PVCs?



- A. DLCI 16: 192.168.10.1/24  
DLCI 17: 192.168.10.1/24  
DLCI 99: 192.168.10.2/24  
DLCI 28: 192.168.10.3/24
  - B. DLCI 16: 192.168.10.1/24  
DLCI 17: 192.168.11.1/24  
DLCI 99: 192.168.12.1/24  
DLCI 28: 192.168.13.1/24
  - C. DLCI 16: 192.168.10.1/24  
DLCI 17: 192.168.11.1/24  
DLCI 99: 192.168.10.2/24  
DLCI 28: 192.168.11.2/24
  - D. DLCI 16: 192.168.10.1/24  
DLCI 17: 192.168.10.2/24  
DLCI 99: 192.168.10.3/24  
DLCI 28: 192.168.10.4/24
29. Which function of the IP SLAs ICMP jitter operation can you use to determine whether a VoIP issue is caused by excessive end-to-end time?
- A. packet loss
  - B. jitter
  - C. successive packet loss
  - D. round-trip time latency
30. Which command enables RSTP on a switch?
- A. spanning-tree uplinkfast
  - B. spanning-tree mode rapid-pvst
  - C. spanning-tree backbonefast
  - D. spanning-tree mode mst

31. What is the function of the command `switchport trunk native vlan 999` on a Cisco Catalyst switch?
- A. It creates a VLAN 999 interface
  - B. It designates VLAN 999 for untagged traffic.
  - C. It blocks VLAN 999 traffic from passing on the trunk.
  - D. It designates VLAN 999 as the default for all unknown tagged traffic.
32. What is the purpose of Inverse ARP?
- A. to map a known IP address to a MAC address
  - B. to map a known DLCI to a MAC address
  - C. to map a known MAC address to an IP address
  - D. to map a known DLCI to an IP address
33. Which command sequence can you enter to create VLAN 20 and assign it to an interface on a switch?
- A. `Switch(config)#vlan 20`  
`Switch(config)#Interface gig x/y`  
`Switch(config-if)#switchport access vlan 20`
  - B. `Switch(config)#Interface gig x/y`  
`Switch(config-if)#vlan 20`  
`Switch(config-vlan)#switchport access vlan 20`
  - C. `Switch(config)#vlan 20`  
`Switch(config)#Interface vlan 20`  
`Switch(config-if)#switchport trunk native vlan 20`
  - D. `Switch(config)#vlan 20`  
`Switch(config)#Interface vlan 20`  
`Switch(config-if)#switchport access vlan 20`
34. On which options are standard access lists based?
- A. destination address and wildcard mask
  - B. destination address and subnet mask
  - C. source address and subnet mask
  - D. source address and wildcard mask

35. Refer to the exhibit. A problem with network connectivity has been observed. It is suspected that the cable connected to switch port Fa0/9 on Switch 1 is disconnected. What would be an effect of this cable being disconnected?



- A. Host B would not be able to access the server in VLAN9 until the cable is reconnected
- B. Communication between VLAN3 and the other VLANs would be disabled
- C. The transfer of files from Host B to the server in VLAN9 would be significantly slower.
- D. For less than a minute, Host B would not be able to access the server in VLAN9. Then normal network function would resume.
36. A network administrator is troubleshooting an EIGRP problem on a router and needs to confirm the IP addresses of the devices with which the router has established adjacency. The retransmit interval and the queue counts for the adjacent routers also need to be checked. What command will display the required information?
- A. Router#show ip eigrp adjacency
- B. Router#show ip eigrp topology
- C. Router#show ip eigrp interfaces
- D. Router#show ip eigrp neighbors
37. At which layer of the OSI model is RSTP used to prevent loops?
- A. physical
- B. data link
- C. network
- D. transport
38. Which encapsulation type is a Frame Relay encapsulation type that is supported by Cisco routers?
- A. IETF
- B. ANSI Annex D
- C. Q9333-A Annex A
- D. HDLC

39. In GLBP, which router will respond to client ARP requests?
- A. The active virtual gateway will reply with one of four possible virtual MAC addresses.
  - B. All GLBP member routers will reply in round-robin fashion
  - C. The active virtual gateway will reply with its own hardware MAC address.
  - D. The GLBP member routers will reply with one of four possible burned in hardware addresses.
40. Which statement describes the process identifier that is used in the command to configure OSPF on a router?
- Router(config)#router ospf 1
- A. All OSPF routers in an area must have the same process ID.
  - B. Only one process number can be used on the same router.
  - C. The process number can be any number from 1 to 65,535.
  - D. Hello packets are sent to each neighbor to determine the process identifier.
41. What OSPF command, when configured, will include all interfaces into area 0?
- A. network 0.0.0.0 255.255.255.255 area 0
  - B. network 0.0.0.0 0.0.0.0 area 0
  - C. network 255.255.255.255 0.0.0.0 area 0
  - D. network all-interfaces area 0
42. If all OSPF routers in a single area are configured with the same priority value, what value does a router use for the OSPF router ID in the absence of a loopback interface?
- A. the IP address of the first Fast Ethernet interface
  - B. the IP address of the console management interface
  - C. the highest IP address among its active interfaces
  - D. the lowest IP address among its active interfaces
43. A router has learned three possible routes that could be used to reach a destination network. One route is from EIGRP and has a composite metric of 20514560. Another route is from OSPF with a metric of 782. The last is from RIPv2 and has a metric of 4. Which route or routes will the router install in the routing table?
- A. the OSPF route
  - B. the EIGRP route
  - C. the RIPv2 route
  - D. all three routes
44. In a GLBP network, who is responsible for the ARP request?
- A. AVF
  - B. AVG
  - C. Active Router
  - D. Standby Router

45. A switch is configured with all ports assigned to VLAN 2 with full duplex FastEthernet to segment existing departmental traffic. What is the effect of adding switch ports to a new VLAN on the switch?
- A. More collision domains will be created
  - B. IP address utilization will be more efficient
  - C. More bandwidth will be required than was needed previously.
  - D. An additional broadcast domain will be created.
46. On a corporate network, hosts on the same VLAN can communicate with each other, but they are unable to communicate with hosts on different VLANs. What is needed to allow communication between the VLANs?
- A. a router with subinterfaces configured on the physical interface that is connected to the switch
  - B. a router with an IP address on the physical interface connect to the switch
  - C. a switch with an access link that is configured between the switches
  - D. a switch with a trunk link that is configured between the switches
47. Which statement about VLAN operation on Cisco Catalyst switches is true?
- A. When a packet is received from an 802.1Q trunk, the VLAN ID can be determined from the source MAC address and the MAC address table.
  - B. Unknown unicast frames are retransmitted only to the ports that belong to the same VLAN
  - C. Broadcast and multicast frames are retransmitted to ports that are configured on different VLAN.
  - D. Ports between switches should be configured in access mode so that VLANs can span across the ports.
48. Refer to the exhibit. Given the output for this command, if the router ID has *not* been manually set, what router ID will OSPF use for this router?

| RouterD# show ip interface brief |              |     |        |        |          |
|----------------------------------|--------------|-----|--------|--------|----------|
| Interface                        | IP-Address   | OK? | Method | Status | Protocol |
| FastEthernet0/0                  | 192.168.5.3  | YES | manual | up     | up       |
| FastEthernet0/1                  | 10.1.1.2     | YES | manual | up     | up       |
| Loopback0                        | 172.16.5.1   | YES | NVRAM  | up     | up       |
| Loopback1                        | 10.154.154.1 | YES | NVRAM  | up     | up       |

- A. 10.1.1.2
- B. 10.154.154.1
- C. 172.16.5.1
- D. 192.168.5.3

49. Which feature can you implement to reserve bandwidth for VoIP calls across the call path?
- A. PQ
  - B. CBWFQ
  - C. round robin
  - D. RSVP
50. What should be part of a comprehensive network security plan?
- A. Allow users to develop their own approach to network security.
  - B. Physically secure network equipment from potential access by unauthorized individuals.
  - C. Encourage users to use personal information in their passwords to minimize the likelihood of passwords being forgotten.
  - D. Delay deployment of software patches and updates until their effect on end-user equipment is well known and widely reported.